

Career-City Fit: Predicting Occupation & Location based Affordability in U.S. Cities

1. Group Members & Duties

Omar El Nesr

Rachel Guhathakurta

We plan to both equally contribute to all parts of the project, including:

- Sourcing datasets will be split equally
- Datasets for cleaning & validation will be equally divided
- Thinking of and implementing exploratory data analysis will be done jointly
- Designing the dashboard will be done by Rachel
- Designing and training the model(s) will be jointly developed

2. Data Sources

Large, public, structured datasets from official U.S. government and labor-market sources. We ultimately want **occupation x metropolitan area x year** level information:

1. **BLS Occupational Employment and Wage Statistics (OEWS)**
The primary source, providing occupation-specific wages and employment metrics, across many occupations and metropolitan areas. ([Bureau of Labor Statistics](#))
2. **U.S. Census American Community Survey (ACS)**
Provides metro-level variables for local socioeconomic and housing context: median rent, household income, education levels, poverty rates, population characteristics. ([Census.gov](#))

Other sources include:

3. **BEA Regional Price Parities (RPP)**
To adjust and account for cost-of-living differences across areas. ([Bureau of Economic Analysis](#))
4. **BLS Local Area Unemployment Statistics (LAUS)**
Captures metro-level labor-market conditions, including unemployment, employment, and labor-force measures. ([Bureau of Labor Statistics](#))

5. **O*NET occupation metadata**
Includes job zone, education expectations, work context, and skill characteristics information. ([O*NET OnLine](#))
 6. **HUD Fair Market Rents (FMR)**
Housing-cost benchmark across many locations. ([HUD Data](#))
 7. **BLS Quarterly Census of Employment and Wages (QCEW)**
Metro-level economic momentum features such as wage growth and employment growth. ([Bureau of Labor Statistics](#))
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3. Objective & Value Proposition

An important question nearly everyone needs to ask is:

“For my career, which cities are actually good places to live and work?”

We would like to make a decision support dashboard that aggregates information on occupation-specific wages, local housing costs, local price levels, labor-market conditions, unemployment levels, and so on that differentiates between occupation and city. The dashboard will allow a user to select a career, explore different regions, see which locations appear strongest on the factors that matter most to them.

We will estimate and deliver information not just on current statistics but assessments and predictions on how strong a metro is right now for a specific occupation, how that metro has changed over time, and where it is likely to head in the future. This has a real practical use case for students, graduates, early professionals, and anyone considering relocation as a dashboard that helps answer a real decision problem instead of just displaying statistics.

4. Modeling Plan

Using the primary two datasets of occupational employment and wage statistics, and American community census, we will be able to construct a combined dataset on the **occupation x metropolitan area x year** level. Each row would represent one occupation, in one metropolitan area, in one year, with features drawn from all of the contextual data. The primary features would be drawn from the BLS and ACS data sources:

- median annual wage,
- percentile wages,
- employment level,
- employment concentration/location quotient,
- lagged wage growth,
- lagged occupation employment growth,
- median rent,
- household income,
- education levels,
- poverty rates

With additional features possible if the other datasets are integrated into the overall table.

With these features, we have two primary targets: a regression score for each location on next-year occupation specific affordability or “opportunity”, and a classification target indicating whether a metro-occupation pair is expected to improve in a meaningful metric in the next year.

There are several model families we can try:

Baselines

- persistence baseline using the previous year’s value,
- rolling historical averages,
- occupation-only and metro-only average baselines.

Interpretable models

- linear regression,
- ridge/lasso regression,
- logistic regression for classification.

Nonlinear models

- random forest,
- gradient-boosted trees.

The main questions we're aiming to be able to answer from EDA and predictive modeling are:

- which occupations have the widest metro-to-metro wage dispersion,
- which metros offer high nominal pay but poor real affordability,
- which metros are balanced on both affordability and opportunity,
- which careers are geographically concentrated versus geographically portable,
- how affordability and opportunity have changed over time,
- and which metros appear to be rising or declining for selected occupations.

5. Anticipated Challenges

We anticipate encountering difficulties on the different specificities and quality of data among and between regions: different sources may use different geographic levels. Some are metro-based, while others may be county-based or use slightly different area definitions. Additionally, not every combination of region-occupation will be equally as detailed or as reliably covered.

There is also the possibility of having “too much” data, in that the scope of the project grows too large and is no longer as focused on the readily available and higher-quality data that is accessible.